

IHCI 22

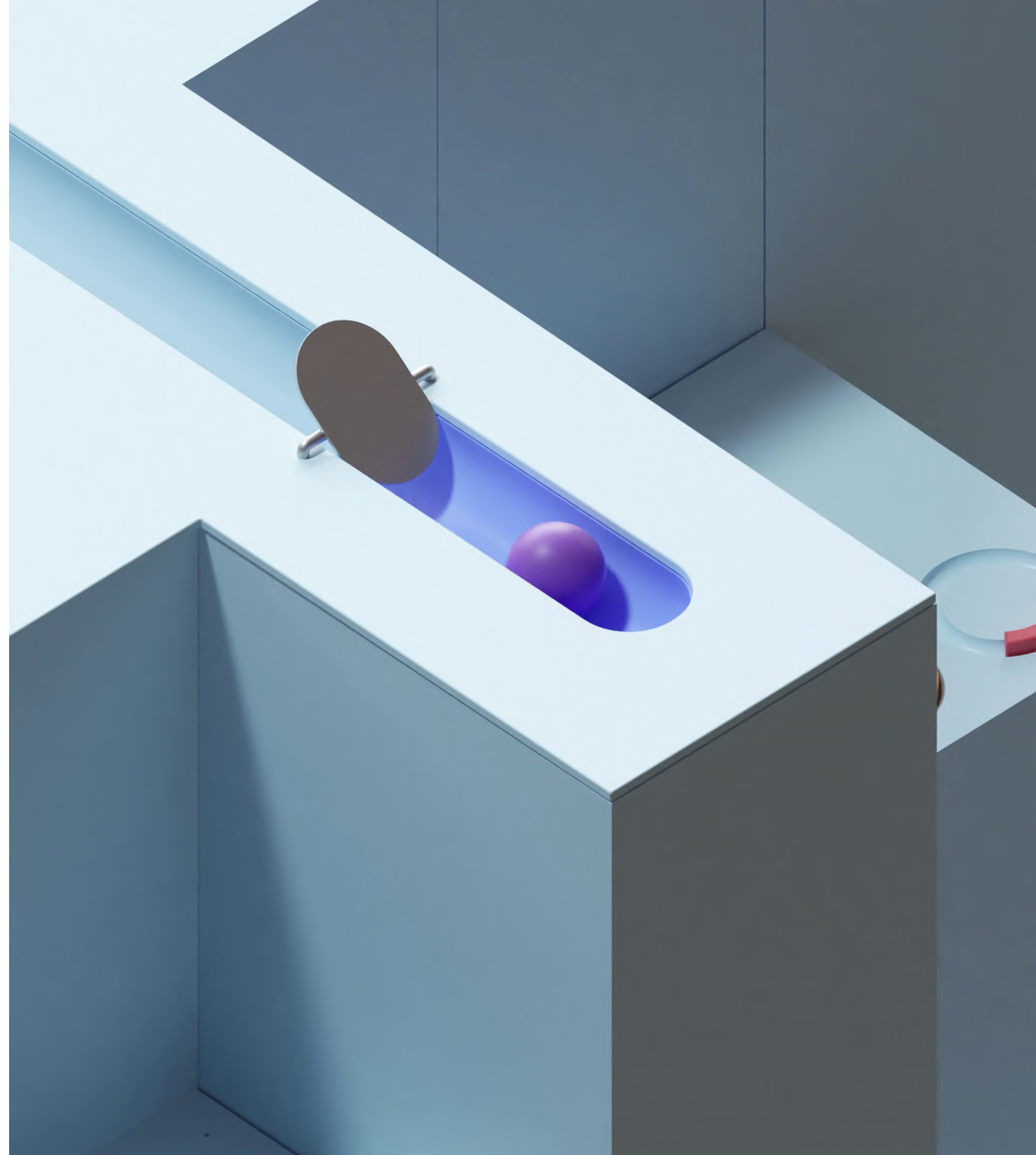
Tutorial

Evaluation & Testing.



Evaluation is integral to the design process.

Collection and analysis of data of the users' and potential users' experiences while interacting with a design artefact such as a prototype or an app is called Evaluation.



Why evaluation?

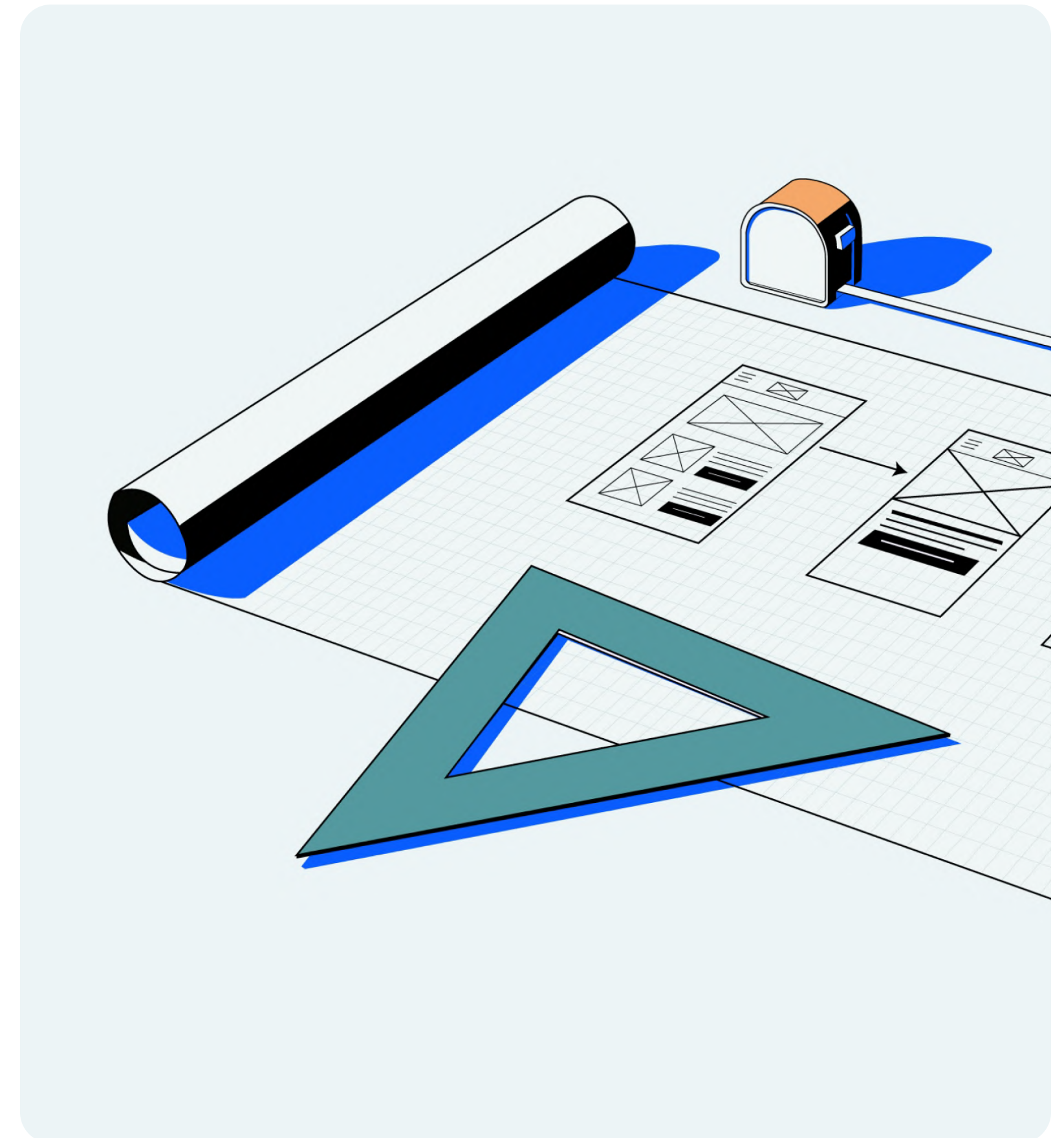
Why do we need to evaluate?

1. Making sure of a pleasing and engaging system

2. Ensuring well designed products sell

3. Gathering varied perspectives

4. Validating the solution statement



Types of Evaluation

Controlled settings directly involving users

These include settings like usability and research labs. Users' activities are controlled to test hypotheses and measure or observe certain behaviours. The main methods are **usability testing** and **experiments**.

Natural Settings directly involving users

There is little to no control of users' activities to determine how the product would be used in the real world. Examples of this is **field / in the wild studies**.

Settings not directly involving the users

Consultants and researchers critique, predict and model aspects of the interface to identify the most obvious usability problems. The range of methods includes inspections, heuristics, walk throughs, models and analysis.

What is usability testing ?

- How usable is a product?
- Traditionally tested in control lab settings
- Helps in finding out design flaws that are generally overlooked.
- Designers have the freedom to make the environment very controlled and influenced using different factors.
- There are also remote or guerrilla settings to conduct usability testing, where designers/ evaluators don't have that much freedom.

Usability Testing Methods

In-person

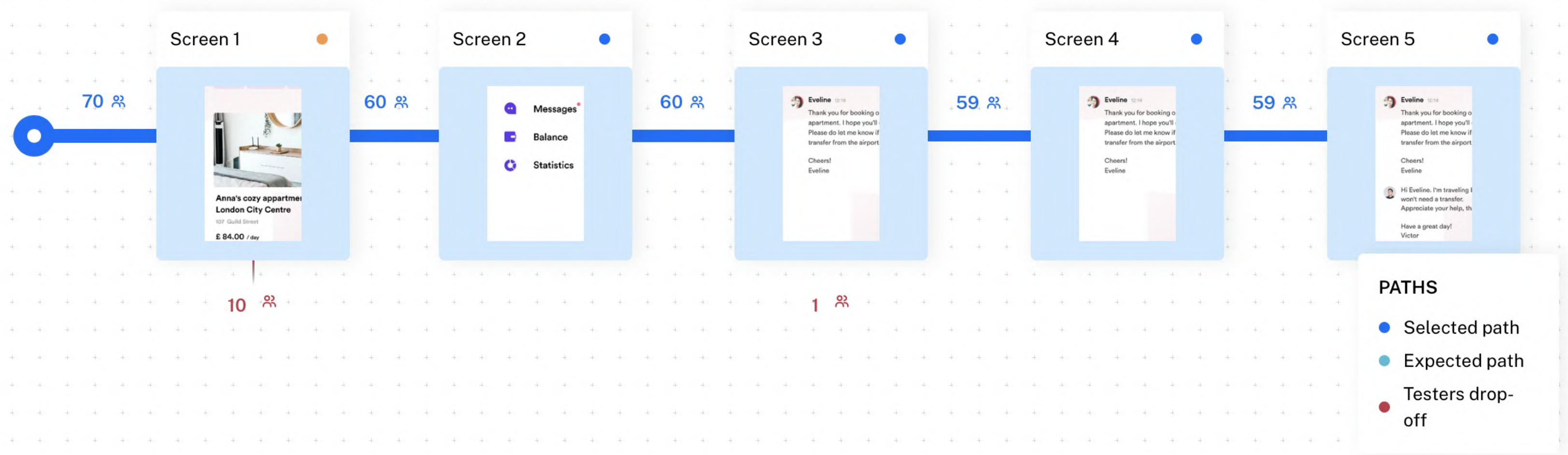
Formal, live testing of representative users requires an empathetic moderator to note testers' experiences.

Remote

Catching users in their own environments can reveal more-accurate "field" insights.

Guerrilla

Testing your design informally on passers-by/colleagues; risks include inaccurate data.



Experiments

Can be used to confirm hypotheses definitively.

- Hypotheses consist of independent and dependent variables.
- Researchers manipulate the independent variables to observe the effects on the independent variables.

In-the-wild / field study

Evaluating designs in natural environments.



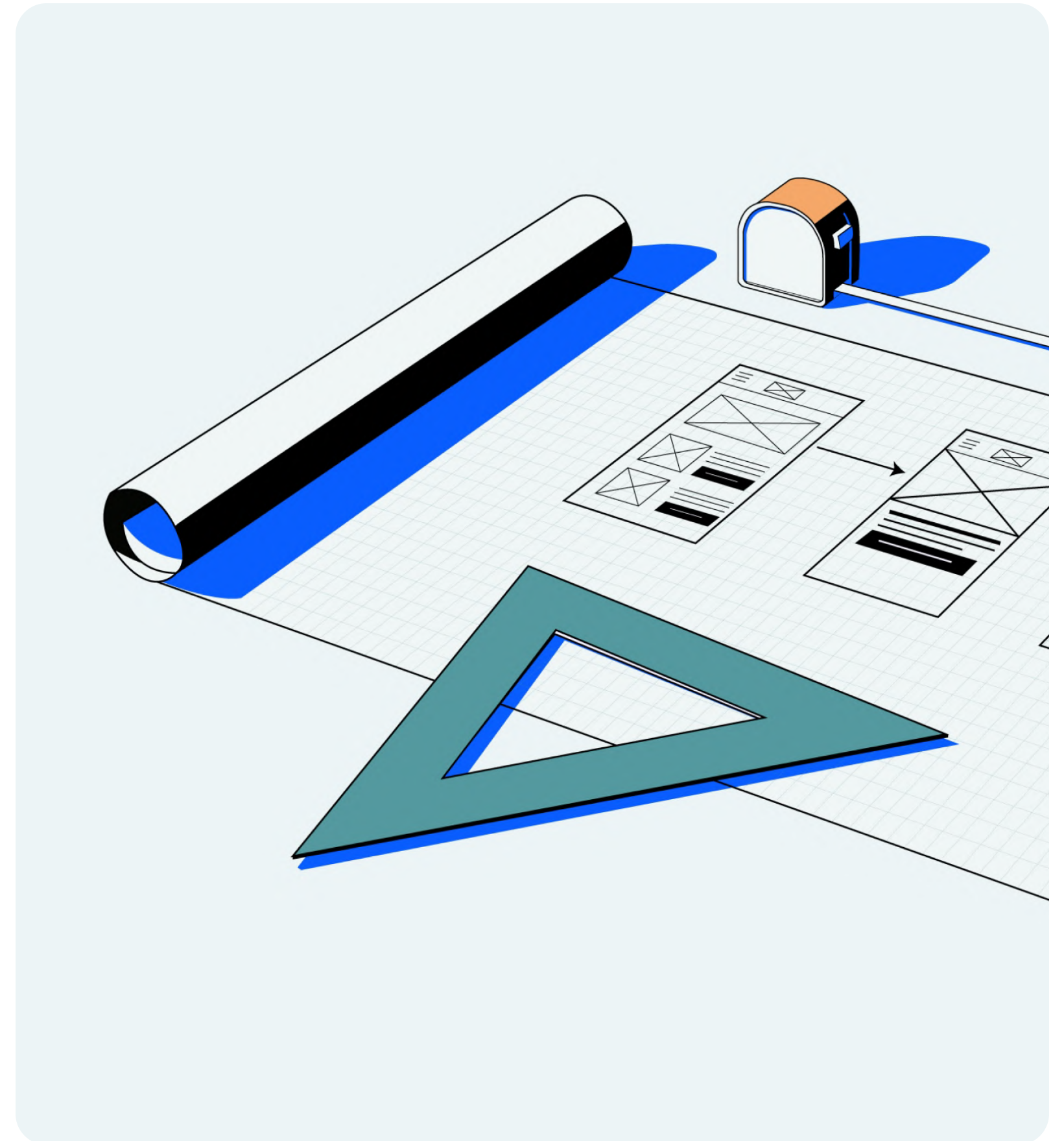
Instead of developing solutions that fit in with existing practices and settings, researchers often explore new technological possibilities that can change and even disrupt participants' behaviour. Opportunities are created, interventions are installed, and different ways of behaving are encouraged. A key concern is how people react, change, and integrate the technology into their everyday lives.

A fast method of field testing is **Guerrilla** testing.

Example of Evaluation

Ritvik Kar & team

https://docs.google.com/presentation/d/1K3EGfZA6i7kOsHCKP0CkaKPIpG8_quPYGglhFY6XcVg/edit#slide=id.g77ff08fbe9_0_0





Thank you

